



E-Commerce Revenue & Customer Satisfaction Analysis

SECTOR

Retail /
E-Commerce

PERIOD

Jan-Mar 2024

DATASET

10K rows
12 columns

Capstone Project 2 | Retail Sector | India | Q1 2024

₹74.84 Cr

Revenue

10,000

Orders

3.01/5

Avg Rating

5.0 Days

Avg Delivery

Tools: Python • Tableau Public • GitHub

Presentation Agenda

01

Business Problem & Dataset Overview

02

ETL Pipeline & Data Cleaning

03

Exploratory Data Analysis

04

Statistical Analysis & Findings

05

Tableau Dashboard Walkthrough

06

Key Insights & Business Recommendations

Business Problem Statement

"Which product categories and cities drive the highest revenue, and what operational factors most influence customer satisfaction ratings across Indian e-commerce?"

Why This Matters

- Category mix decisions impact marketing budget allocation
- City performance informs regional expansion plans
- Customer satisfaction drives retention and LTV

Dataset Snapshot

Rows:	10,000 transactions
Columns:	12 (8 analytical)
Period:	Jan – Mar 2024
Cities:	Mumbai, Delhi, Pune, Chennai, Bangalore
Categories:	Clothing, Electronics, Furniture, Groceries

Data Cleaning & ETL Pipeline

01

Extract

Load raw CSV
10,000 rows × 12 cols

02

Normalise

Fix city casing
'pune' → 'Pune' (200 rows)

03

Impute

Fill nulls: product_name,
payment, delivery (500
each)

04

Engineer

Add: month, fast_delivery,
high_value, high_rating flags

05

Load

Save to data/processed/
Tableau-ready CSVs

Data Quality Issues Found & Resolved

- product_name: 500 nulls → Filled 'Unknown'
- payment_method: 500 nulls → Filled 'Unknown'

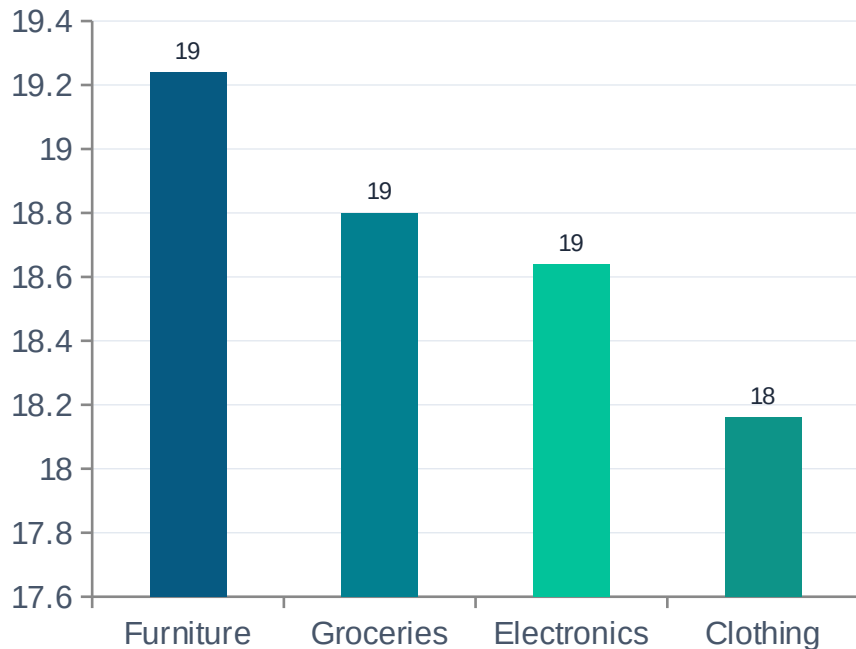
- delivery_days: 500 nulls → Filled 'Unknown'
- city: 200 rows lowercase → Normalised

✓ Output

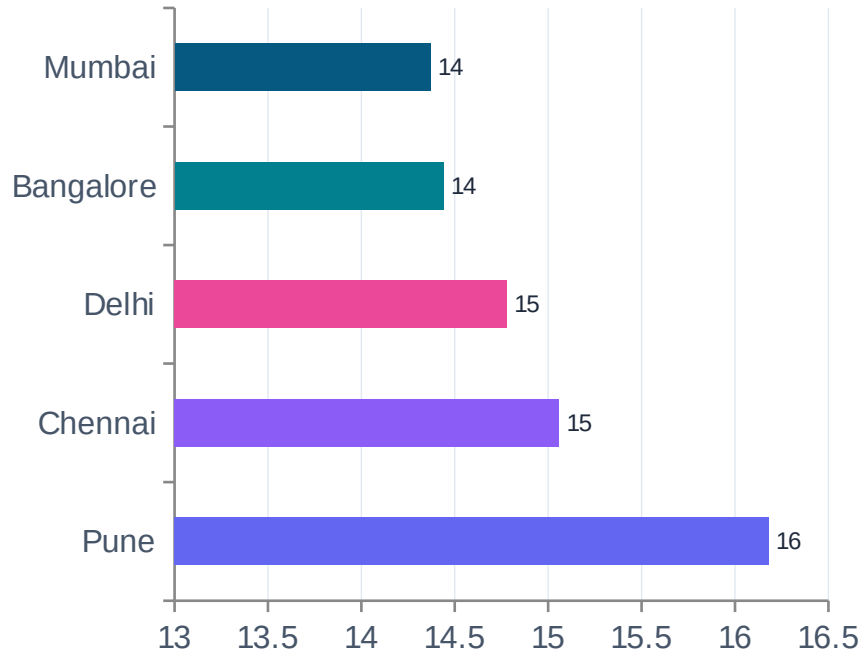
10,000 rows
16 columns
0 nulls remaining

Revenue Analysis — Categories & Cities

Revenue by Category (₹ Crore)



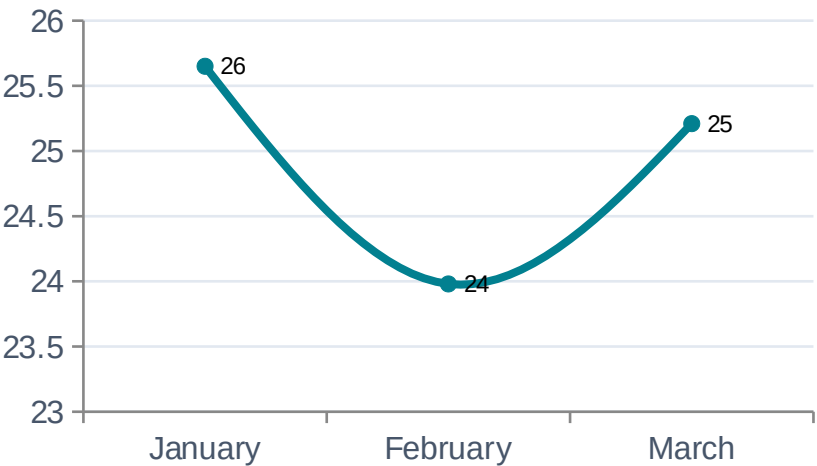
Revenue by City (₹ Crore)



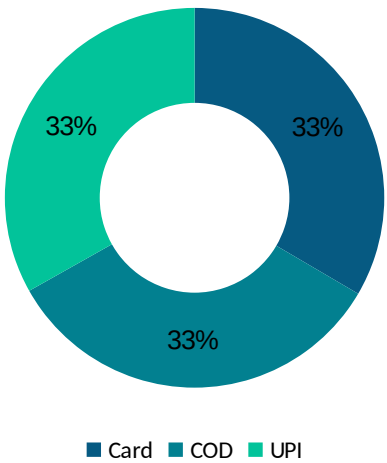
Key Insight: All categories within 1.4% of each other — no dominant category. Pune leads cities by 8.1%.

Trends, Payments & Customer Ratings

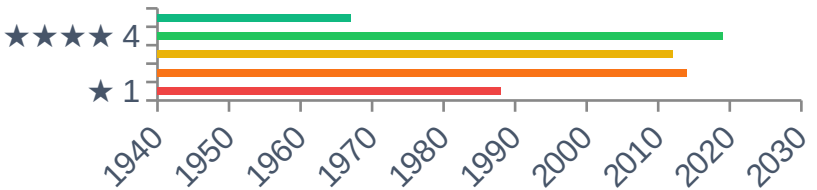
Monthly Revenue Trend



Payment Method Split



Customer Rating Distribution



Key Insights

- January peaks at ₹25.65 Cr (+7% vs avg)
- UPI + Card = 63.3% of transactions
- Ratings are nearly uniform (1988–2019 orders each)
- Avg rating: 3.01/5 — significant improvement opportunity

Statistical Analysis Results

Pearson Correlation

Delivery Days → Rating

$r = -0.007$

$p = 0.471$

Not Significant

One-Way ANOVA

Rating across Categories

$F = 1.462$

$p = 0.223$

Not Significant

Linear Regression

Delivery Days → Rating

$R^2 \approx 0.0001$

$p = 0.471$

No Predictive Power

Chi-Square Test

Payment Method vs High-Value

$\chi^2 = 1.8$

$p = 0.402$

No Association

T-Test

Pune AOV vs Other Cities

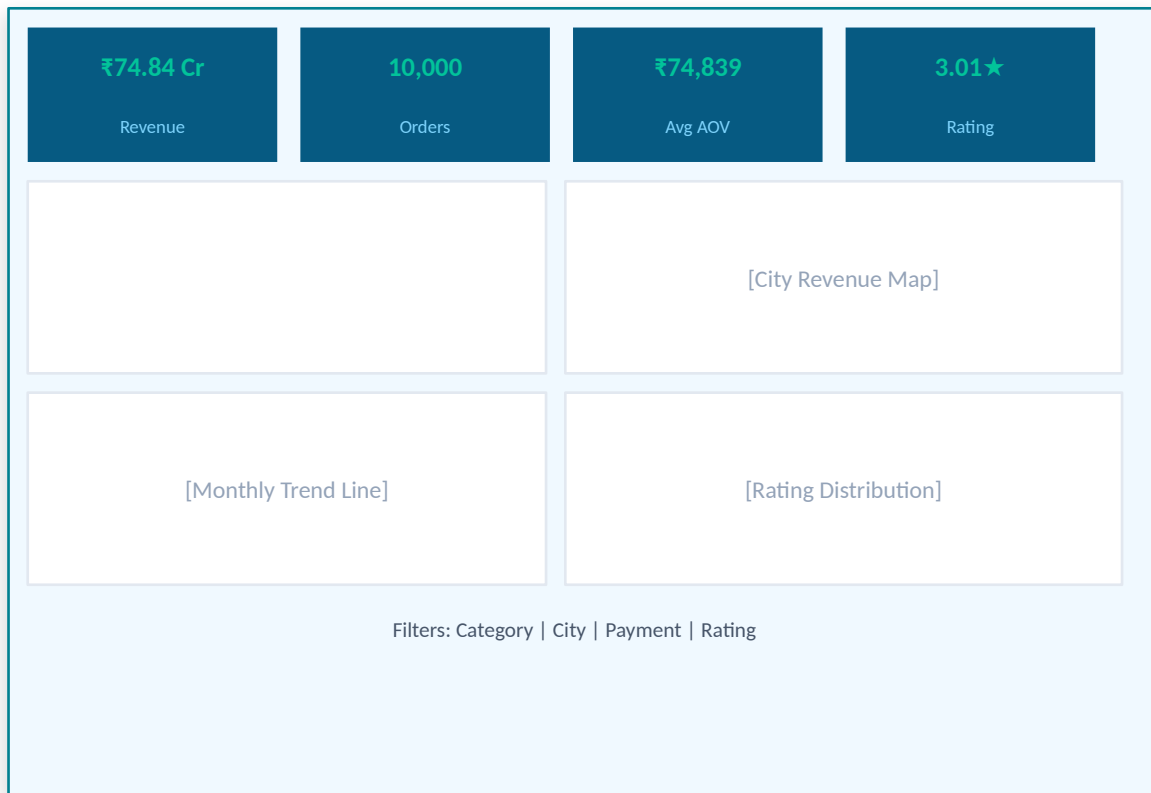
$t = 0.92$

$p = 0.358$

Not Significant

Core Finding: No operational variable significantly predicts customer ratings — satisfaction is driven by product & service quality, not logistics.

Tableau Dashboard Overview



Dashboard Features

- ✓ 5 KPI summary cards
- ✓ Monthly revenue trend (line chart)
- ✓ Revenue by category (doughnut)
- ✓ Revenue by city (bar chart)
- ✓ Rating distribution (bar chart)
- ✓ Payment method split (doughnut)
- ✓ Category performance table
- ✓ 4 interactive filter controls
- ✓ Insight annotation panels

8 Key Business Insights

1 Category Revenue Parity

All 4 categories: 24–26% share. No dominant category.

2 Pune Is the #1 Market

Leads at ₹16.18 Cr — 8.1% above city average.

3 Satisfaction Universally Low

3.01/5 avg with uniform 1–5 distribution. Systemic issue.

4 January Revenue Peak

₹25.65 Cr — 7% above Q1 average. Seasonal pattern.

5 Digital Payments Lead

UPI + Card = 63.3%. COD declining toward margin.

6 Delivery ≠ Satisfaction

$r = -0.007$. Faster delivery won't fix ratings.

7 High-Value Orders Spread

Top 10% orders equally distributed. No category premium.

8 5% Data Gap

500 orders each missing payment & delivery data.

5 Business Recommendations

REC 1

HIGH

Launch Category Premium Tiers

Introduce premium product tiers across all categories to break revenue parity and drive higher-margin orders.

↑ AOV by 15-25%

REC 2

HIGH

Pune Loyalty Programme

City-specific loyalty programme with double points and early sale access to leverage Pune's revenue lead.

↑ Retention by 20%

REC 3

CRITICAL

Fix Satisfaction — Product Quality Audits

Delivery speed is irrelevant. Focus on product quality audits (Clothing, Furniture lowest rated) and post-purchase service.

↑ Rating to 3.5+

REC 4

MEDIUM

Incentivise Digital Payments

Cashback for UPI/Card payments. Shifting 10% from COD saves ₹15-25 per order in handling costs.

↓ COD cost ₹15-25/order

REC 5

MEDIUM

Mandatory Data Completeness Checks

Real-time validation at checkout for payment_method (required field) and delivery confirmation triggers.

100% data coverage

₹74.84 Cr

Total Revenue

10,000

Orders Analysed

5 Cities

Coverage

5 Recs

Delivered

Thank You

Questions & Discussion

Key Takeaway

Customer satisfaction (3.01/5) is the
#1 business risk to fix — not logistics.

Python + Jupyter

Tableau Public

GitHub

Statistical Analysis

"Tools don't get you marks. Thinking gets you marks."